INCH-POUND

A-A-58049
20 OCTOBER 1995
SUPERSEDING
MIL-M-6867,TYPE II
27 January 1969

COMMERCIAL ITEM DESCRIPTION

MAGNETIC INSPECTION UNIT, STATIONARY 52 INCH

The General Services Administration has authorized the use of this commercial item description to replace MIL-M-6867, Type II.

1. SCOPE. This commercial item description covers 52 inch wet horizontal magnetic particle inspection unit which shall be capable of performing conventional magnetization of ferromagnetic parts.

2. Salient characteristics:

2.1 It shall provide alternating current (AC) and full wave rectified direct current (DC) from a three phase AC power supply.

2.1.1 The input power 200-240 VAC with 230 VAC nominal or 400-480 VAC with 460 VAC nominal, 50-60 Hertz, 3 phase. The unit shall have a press-to-test system indicating that the proper electrical connections have been made.

2.1.2 The rated maximum output shall be at least DC 6000 amperes with not more than 6% ripple, and for AC is 5000 amperes.

2.1.3 The unit shall be self-contained requiring only connections for electrical and compressed air supply for operation.

2.2 The unit shall be capable of performing magnetic particle inspection processes as described ASTM E1444.

2.3 The unit shall be capable of magnetizing workpieces in the longitudinal and circular directions.

2.4 The maximum contact head opening shall be 52 $\pm$ 2inches.

2.5 The overall length of the unit shall not be greater 73 inches.

2.6. Instruments for inspection shall be conveniently located to permit the operator manipulation and viewing from the inspection station.

2.6.1 A digital ammeter shall display AC root mean square(RMS) amperes or DC amperes at the coil or contact. The meter shall incorporate a sample and hold the display that will be reset after magnetizing or demagnetizing current push button is depressed.

2.6.2 The current control shall be continuously variable.

2.6.3 The magnetization activation switch shall be of a type that can be activated by a knee push bar.

Beneficial comments(recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to San Antonio ALC/TILDD, Bldg. 207 Kelly AFB TX 78241-6425 Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter.

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2.6.4 The magnetization time shall be variable from 0.20 to 2.0 seconds.

2.6.5 The headstock pressurization shall be foot switch activated.

2.6.6 The coil shall be rail mounted, have an inside diameter of 16 inches, and be capable of producing 15,000 Ampere-turns AC and DC.

2.7 The unit shall be capable of demagnetizing any workpiece previously magnetized by the same unit to less than three gausses in any directions. Both reversing DC and AC ramp demagnetization capabilities shall be built into the unit.

2.8 The air pressure operating range shall be minimum of 60 to 110 PSI..

2.9 The equipment shall be capable of operating at the following duty cycles indefinitely without damage to any components

Period magnetizing current "on"	Period magnetizing current "off"	Allowable percent of maximum rated magnetizing current
10 seconds	3 minutes	40 percent
5 seconds	4 minutes	100 percent

2.10 Electrical systems components, devices, wiring and insulation shall be suitable for voltages, current and duty characteristics of the circuits in which they are used shall meet the applicable requirements of the National Electrical Manufactures Association (NEMA) standards and the National Electrical Code (NEC).

2.11. Accessories. The following accessories shall be included:

2.11.1 A copper rod one inch in diameter by eighteen inches in length that will provide a low resistance path for determining magnetizing current through the contact heads.

2.11.2 Four auxiliary cables consisting of two No. 4/0 extra flexible insulated conductors. with each cable being 15 feet in length, and two No. 2/0 extra flexible insulated cable conductors with each being 10 feet in length. Each conductor shall have a 4/0 connector on each end and shall have a suitable heavy duty, oil-resisting insulation.

2.11.3 A 100 Watt black light for viewing fluorescent magnetic particle inspection shall be provided

2.11.3.1 The black light shall have an intensity of at least 1000 microwatts per square centimeter at 15 inches, and shall have a peak transmission wavelength of 365 manometers $\pm$ 1.0%.

2.11.3.2 The lamp shall be mounted on swivel brackets which shall permit the lights to be directed at the parts under inspection.

2.11.4 A hood suitable for fluorescent inspections including flame retardent curtains, a white light source and ventilation fan shall be provided.

2.11.4.1 The hood shall allow not more than 2 foot-candles(20 lux) of ambient light into the inspection area.

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2.11.5 A set of replaceable copper braid pads for the headstock and tailstock contact plates.

2.11.6 The unit shall have a tank suitable for holding the suspension and particles, hoses long enough to reach the entire inspection area, and nonmagnetic low pressure spray nozzles.

2.11.7 The unit shall have an agitation system capable of maintaining magnetic particles in suspension.

2.11.8 The unit shall be delivered with all equipment necessary for immediate operation(e.g. circuit breaker box, air pressure regulator, air filter).

3 Regulatory requirements. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

3.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. Quality assurance provisions.

4.1 CONTRACTOR CERTIFICATION. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, standards, and quality assurance practices. The government reserves the right to require a government witnessed, pre-award, sample verification test at the contractor's site to insure conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

5. Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be specified in the contract or order.

6.0 NOTES

6.1 Address for obtaining reference documents.

ASTM E1444 Standard Practice for Magnetic Particle Examination
American Society for Testing and Materials
100 Barr Harbor Drive
west Conshokocken Pa 19428

6.2 Ordering data.

a) Title, number, and date of this description.
b) Specify special packaging requirements.
c) Contracting officer may request proof of certification of commercial item prior to first contract delivery.

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6.3 National Stock Number.

6635-01-396-2798

MILITARY INTEREST:

CUSTODIANS:
Air Force - 99
Navy - AS
Army - GL
DLA-GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:
AIR FORCE 82

AGENT:
AIR FORCE 99

PROJECT NUMBER: 6635-0159

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